

Comparison of Means with R

Amandine Gamble

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```
# Packages  
library(ggplot2)  
library(dplyr)
```

1 T-test

We will use simulated (fake!) data on daily calorie intake across countries.

1.1 Question and hypotheses

How does calorie intake varies across countries? Specifically, is calorie intake different between Ireland and Japan?

- **H1**: there is a significant association between calorie intake and country.
- **H0**: there is **no** significant association between calorie intake and country.

1.2 Data Simulation

In real life, this would correspond to data collection. Think about how to reach random sampling (i.e. independence of the individuals), how the data should be recorded (continuous, discrete, categorical...) and sample size.

```
# Set the seed for reproducibility  
set.seed(123)  
  
# Number of observations for each country  
num_ireland <- 1000  
num_japan <- 1000  
  
# Generate calorie intake values for Ireland and Japan  
calories_ireland <- rnorm(num_ireland, mean = 3885, sd = 500)  
calories_japan <- rnorm(num_japan, mean = 2705, sd = 400)
```

```
# Create a data_cal frame
data_cal <- data.frame(
  country = c(rep("Ireland", num_ireland), rep("Japan", num_japan)),
  calorie_intake = c(calories_ireland, calories_japan)
)

# Display the first few rows of the simulated data_calset
head(data_cal)
```

1.3 Data Exploration

1.3.1 Figure

```
# Create a histogram with different colors for each country
ggplot(data_cal, aes(x = calorie_intake, fill = country)) +
  geom_histogram(binwidth = 150, position = "dodge", color = "black", alpha = 0.7) +
  labs(x = "Daily calorie intake (Kcal)",
       y = "Individual count") +
  scale_fill_manual(values = c("Ireland" = "#019529", "Japan" = "#BC002D"), name = "Country") +
  theme_minimal()
```

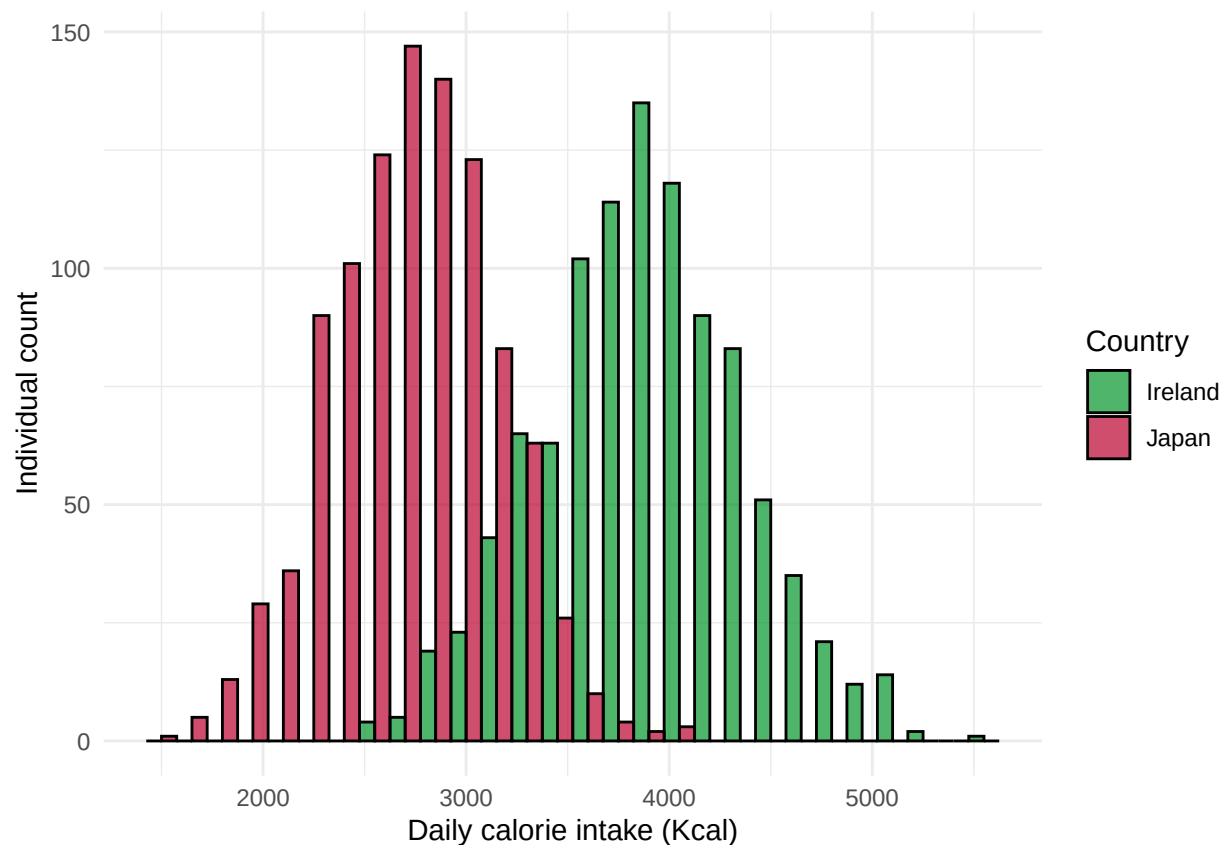


Figure 1: Distribution of daily calory intake.

Visually, calorie intake looks higher in Ireland than in Japan.

1.3.2 Summary Statistics

```
mean(data_cal[data_cal$country=="Ireland", "calorie_intake"])
sd(data_cal[data_cal$country=="Ireland", "calorie_intake"])
nrow(data_cal[data_cal$country=="Ireland",])

mean(data_cal[data_cal$country=="Japan", "calorie_intake"])
sd(data_cal[data_cal$country=="Japan", "calorie_intake"])
nrow(data_cal[data_cal$country=="Japan",])
```

Average (standard deviation) daily calorie intake is 3,893 (496) Kcal in Ireland, and 2,722 (404) in Japan, based on a sample of 1,000 individuals in each country.

1.4 Statistical Test

1.4.1 Assumptions

1.4.1.1 Independance *Observations from each population are independent of one another.* This is likely the case (would be determined by the study design).

1.4.1.2 Normality *Data is normally distributed within each group.* This is visible on the histogram. Make sure the data is stratified by group.

```
# Create a histogram without stratification for comparison
ggplot(data_cal, aes(x = calorie_intake)) +
  geom_histogram(binwidth = 150, color = "black", alpha = 0.7) +
  labs(x = "Daily calorie intake (Kcal)",
       y = "Individual") +
  theme_minimal()
```

This can also be assessed with a qqplot, but they are difficult to interpret.

```
# Create a QQ plot
ggplot(data_cal, aes(sample = calorie_intake)) +
  stat_qq() +
  labs(title = "Whole data set",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```

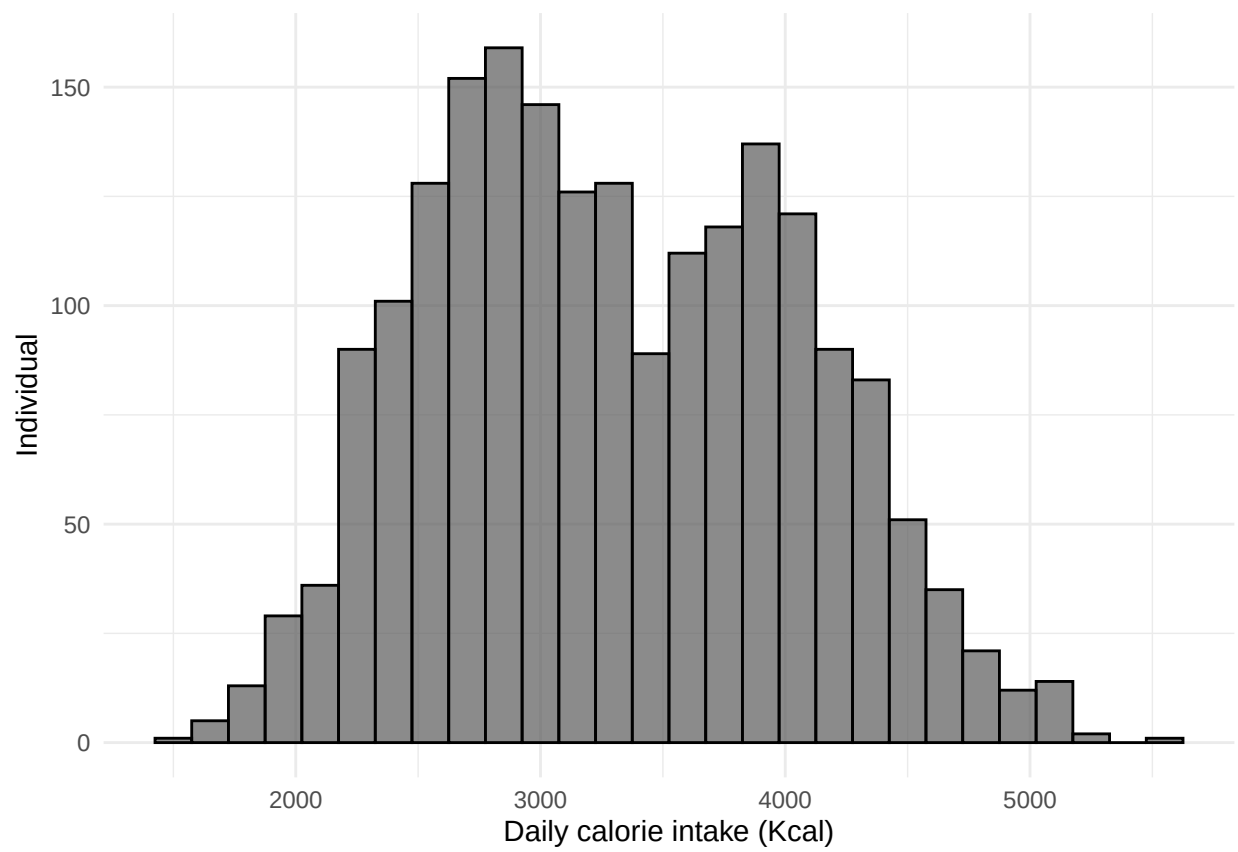
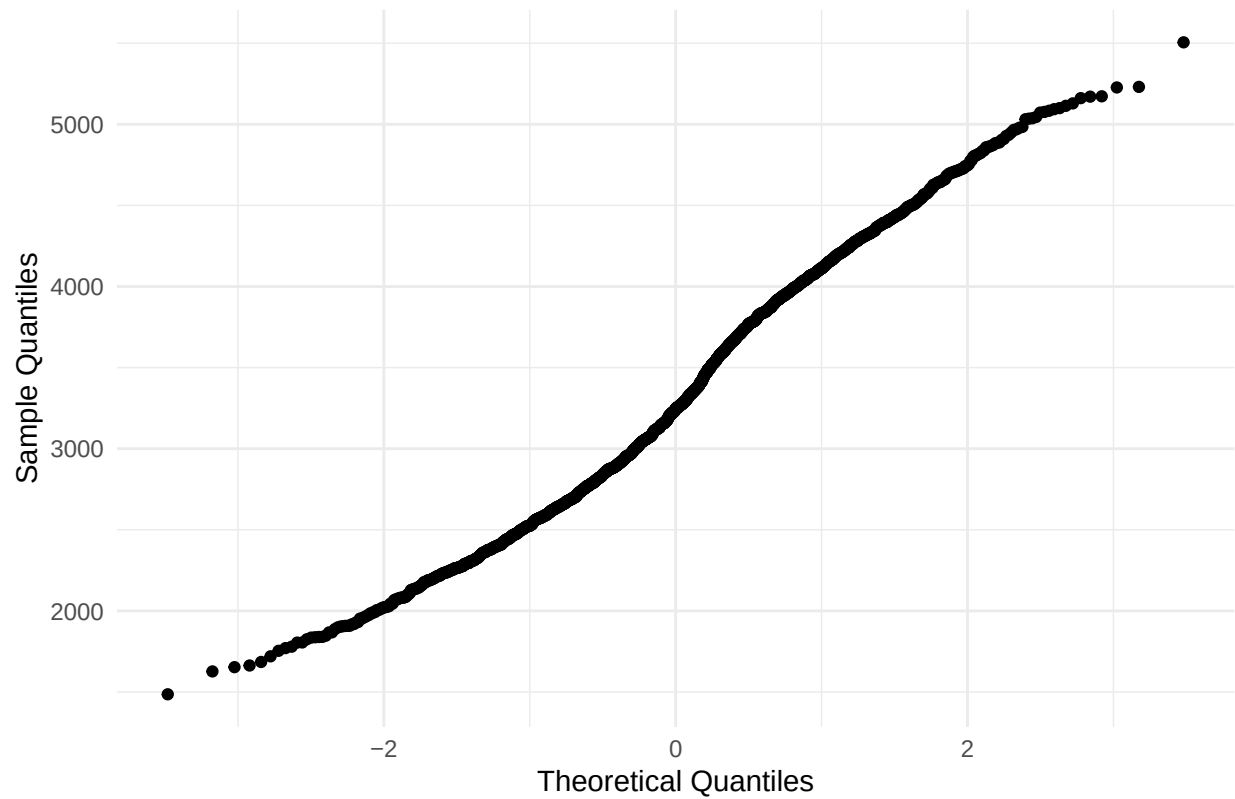
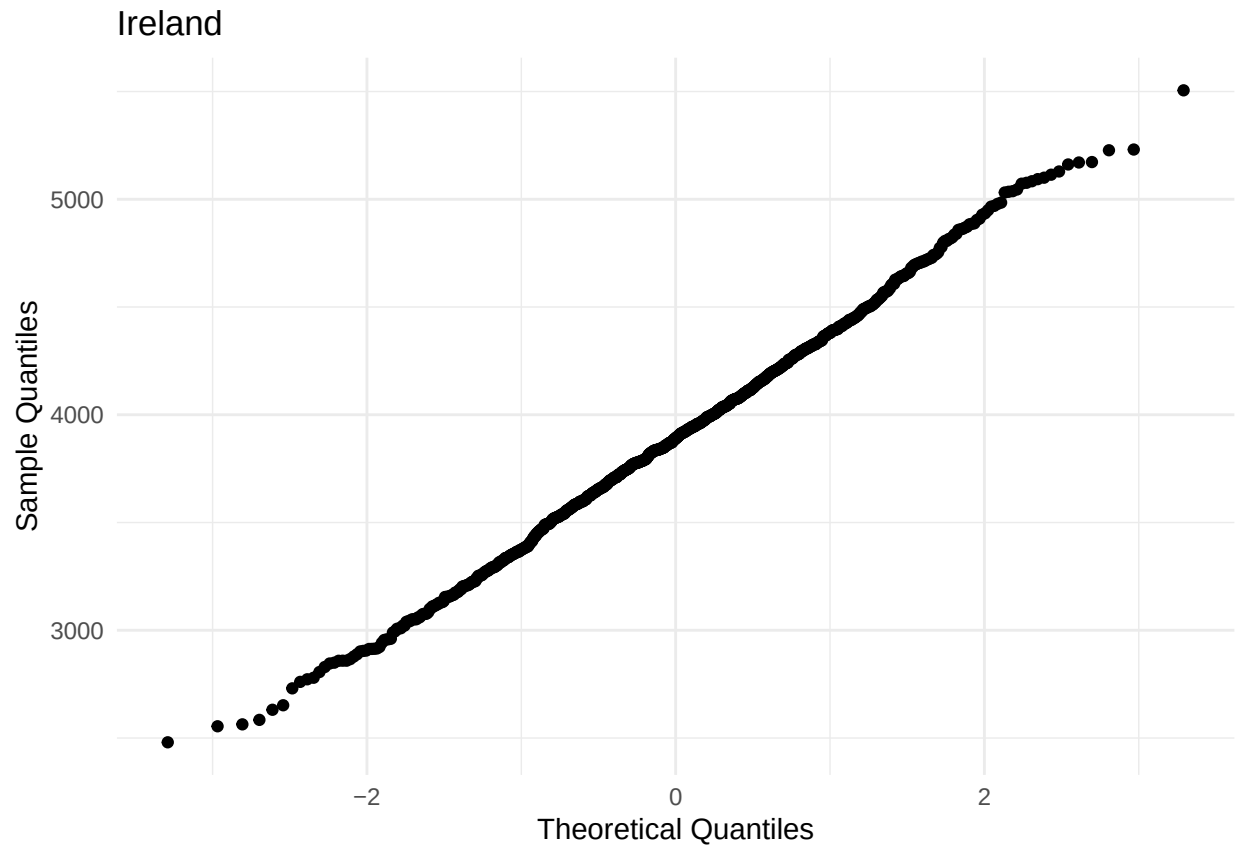


Figure 2: Distribution of daily calorie intake (Kcal).

Whole data set



```
# Ireland only
ggplot(subset(data_cal, country == "Ireland"), aes(sample = calorie_intake)) +
  stat_qq() +
  labs(title = "Ireland",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```



This can also be verified with the Shapiro-Wilk test, **for each group**.

```
# Check normality
shapiro.test(data_cal[data_cal$country=="Ireland", "calorie_intake"])
shapiro.test(data_cal[data_cal$country=="Japan", "calorie_intake"])
```

The observed distributions do not statistically differ from a normal distribution.

1.4.1.3 Homoscedasticity *Variances are equal across populations.* This is visible on the histogram.

This can also be verified with the variance F test test, **for each group**.

```
# Check homoscedasticity
var.test(calorie_intake ~ country, data_cal)
```

The variances are significantly different. However, these tests tend to be very sensitive (strict). I often ignore their output and use the visual assessment instead.

1.4.2 Test

```
# Equal variance t-test
t.test(calorie_intake ~ country, data_cal, var.equal = T)

# Unequal variance t-test (sanity check as the assumption was slightly violated)
t.test(calorie_intake ~ country, data_cal, var.equal = F)
```

We reject the null hypothesis, i.e., we detect a difference. Output fits with expectations.

1.4.3 Explanation

Using a t-test, we detected a significant association between calorie intake and country ($t = 57.908$, $df = 1919.4$, $p\text{-value} < 2.2e-16$). Average (standard deviation) daily calorie intake is higher in Ireland - 3,893 (496) Kcal - than Japan 2,722 (404).

```
# Calculate summary statistics
summary_stats <- data_cal %>%
  group_by(country) %>%
  summarise(
    mean_cal = mean(calorie_intake),
    sd_cal = sd(calorie_intake),
    n = n(),
    se = sd_cal / sqrt(n),
    ci_lower = mean_cal - qt(0.975, df = n-1) * se,
    ci_upper = mean_cal + qt(0.975, df = n-1) * se,
    .groups = 'drop'
  )

# Create the plot
ggplot(data_cal, aes(x = country, y = calorie_intake)) +
  # Raw data points (in background)
  geom_jitter(alpha = 0.3, width = 0.2, size = 0.8, color = "gray50") +
  # Mean points
  geom_point(data = summary_stats, aes(y = mean_cal),
            size = 2, color = "black", shape = 18) +
  # Confidence interval error bars
  geom_errorbar(data = summary_stats,
               aes(y = mean_cal, ymin = ci_lower, ymax = ci_upper),
               width = 0.1, size = 0.5, color = "black") +
  # Labels and theme
  labs(
    x = "Country",
    y = "Daily calorie intake (Kcal)",
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11)
  )
)
```

2 Wilcoxon Rank Test

A non-parametric test would also work (they almost always work!).

```
# Wilcoxon test
wilcox.test(calorie_intake ~ country, data_cal)

##
## Wilcoxon rank sum test with continuity correction
##
## data:  calorie_intake by country
## W = 967234, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

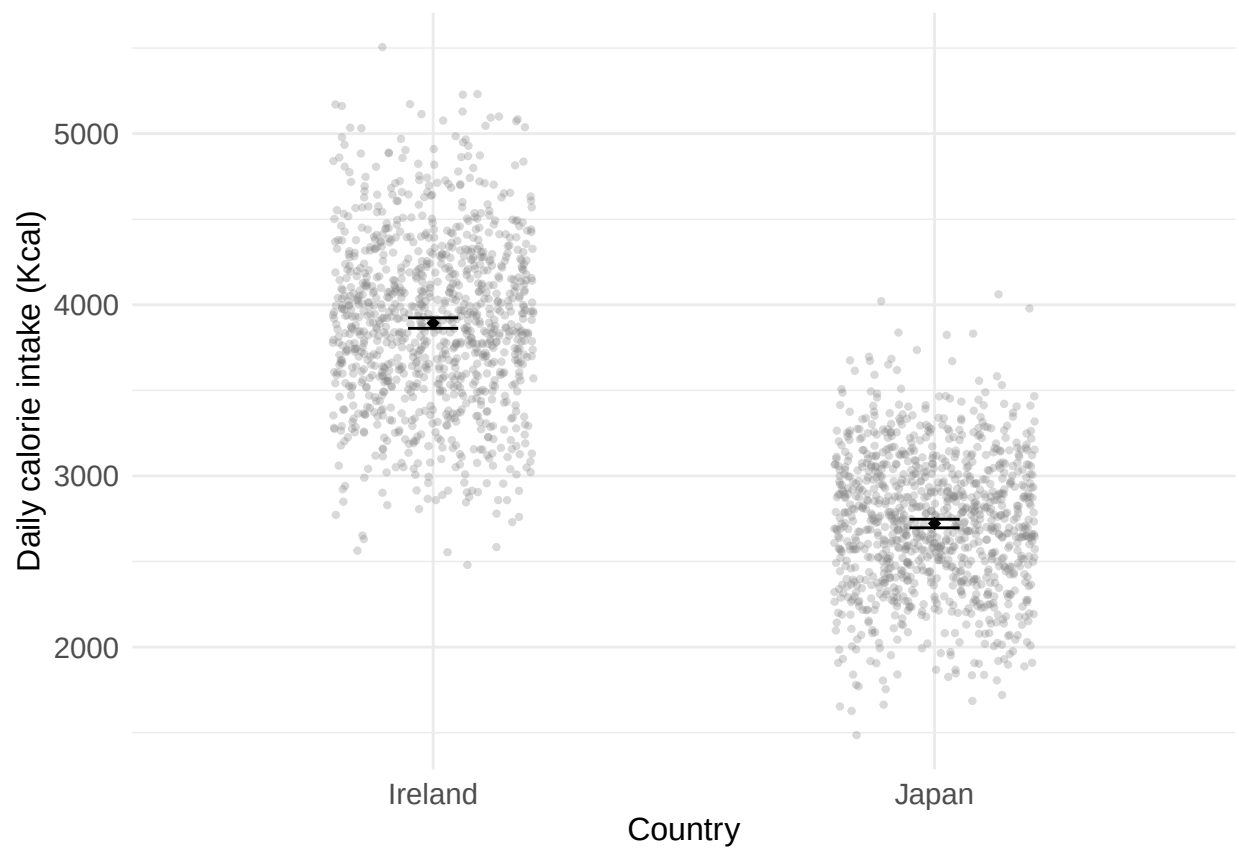


Figure 3: Daily calorie intake (Kcal) in Ireland and Japan. Points represents raw data, and diamond and error bars the mean and 95% confidence interval.

We reach the same conclusion (=

3 One-sample t-test

T-tests can also be used to compare one group of observations to a reference value. For instance, let's consider 2,500 Kcal the recommended daily calorie intake.

```
# Ireland
t.test(data_cal[data_cal$country=="Ireland", "calorie_intake"], mu = 2500)

##
## One Sample t-test
##
## data: data_cal[data_cal$country == "Ireland", "calorie_intake"]
## t = 88.843, df = 999, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 2500
## 95 percent confidence interval:
## 3862.294 3923.834
## sample estimates:
## mean of x
## 3893.064
```

Ireland daily calorie intake is significantly different from the recommended intake ($t = 88.843$, $df = 999$, $p\text{-value} < 2.2e-16$). It is higher.

```
# Japan
t.test(data_cal[data_cal$country=="Japan", "calorie_intake"], mu = 2500)

##
## One Sample t-test
##
## data: data_cal[data_cal$country == "Japan", "calorie_intake"]
## t = 17.381, df = 999, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 2500
## 95 percent confidence interval:
## 2696.924 2747.048
## sample estimates:
## mean of x
## 2721.986
```

Japan daily calorie intake is significantly different from the recommended intake ($t = 17.381$, $df = 999$, $p\text{-value} < 2.2e-16$). It is higher.

4 Activity

Edit the simulation to run the same tests with a sample size of 30 in each group.

5 Paired t-test

We will use simulated (fake!) data on ADHD treatment.

5.1 Question and hypotheses

Is ADHD score improved after treatment? I.e., is there a difference in ADHD score before and after treatment?

- **H1:** there is a significant association between score and treatment.

- **H0**: there is **no** significant association between score and treatment.

5.2 Data Simulation

In real life, this would correspond to data collection. Think about how to reach random sampling (i.e. independence of the individuals), how the data should be recorded (continuous, discrete, categorical...) and sample size.

```
# Set seed for reproducibility
set.seed(123)

# Number of individuals
n_individuals <- 30

# Simulate pre-treatment scores
pre_treatment <- rnorm(n_individuals, mean = 70, sd = 15)

# Simulate post-treatment scores
post_treatment <- rnorm(n_individuals, mean = 60, sd = 15)

# Combine into a single data frame
data_adhd <- data.frame(
  Individual = 1:n_individuals,
  Pre = pre_treatment,
  Post = post_treatment
)

# Display the simulated dataset
head(data_adhd)
```

5.3 Data Exploration

```
# Reshape data for ggplot
data_adhd_long <- tidyr::gather(data_adhd, key = "Treatment", value = "Score", -Individual)
```

5.3.1 Histogram

```
# Create a histogram with different colors for pre- and post-treatment
ggplot(data_adhd_long, aes(x = Score, fill = Treatment)) +
  geom_histogram(position = "identity", alpha = 0.7, bins = 20) +
  labs(x = "Score",
       y = "Frequency") +
  scale_fill_manual(values = c("Pre" = "#65BDBF", "Post" = "#32658A")) +
  facet_grid(Treatment ~ ., scales = "free_y") + # Stack facets vertically
  theme_minimal() +
  theme(legend.position = "none")

#ggsave("histo_raw.pdf", width = 4, height = 4)
```

It shows distribution but not how scores from on given individual relate.

5.3.2 Scatter plot

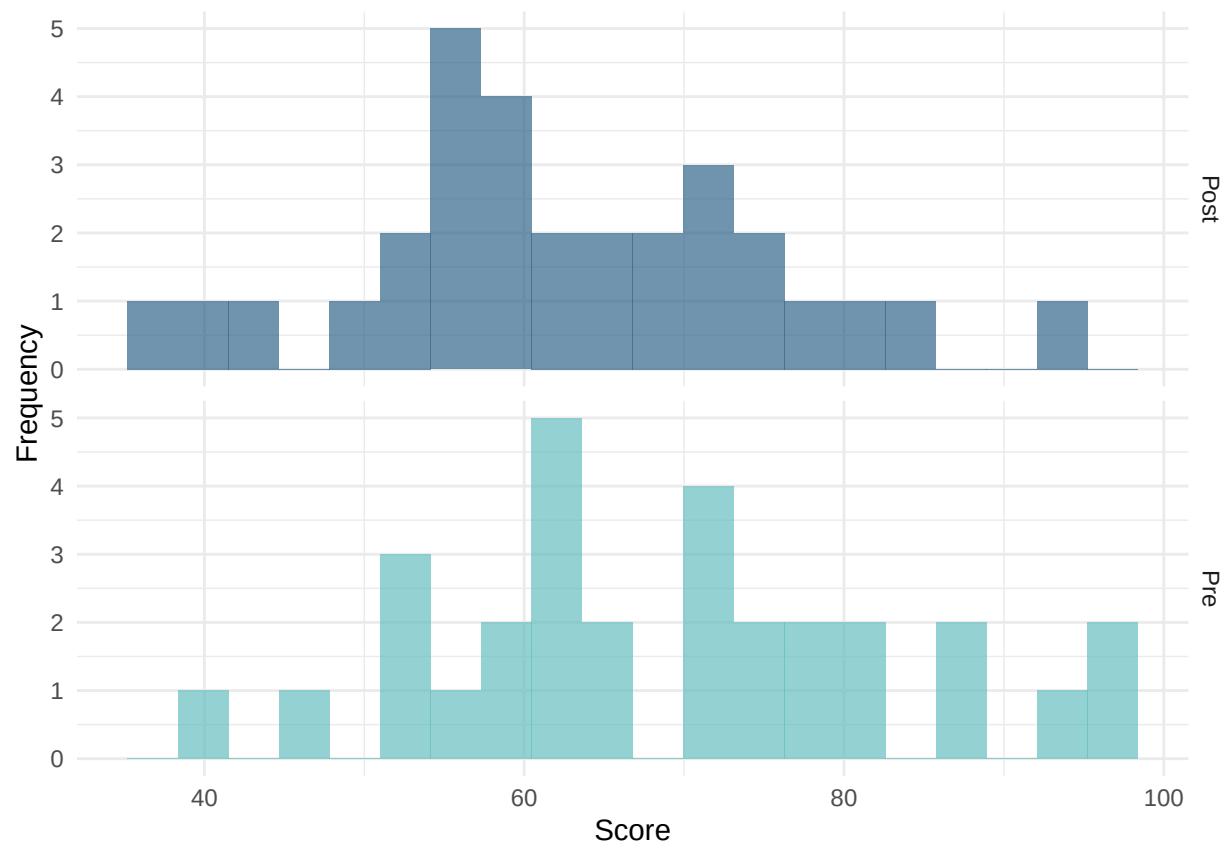


Figure 4: ADHD scores in individuals pre- and post-treatment. Lines connect observations from a given individual.

```
# Create a dot plot with connecting lines
ggplot(data_adhd, aes(x = Individual)) +
  geom_point(aes(y = Pre), color = "#65BDBF", size = 3) +
  geom_point(aes(y = Post), color = "#32658A", size = 3) +
  geom_segment(aes(x = Individual, xend = Individual, y = Pre, yend = Post),
    color = "gray", linetype = "dashed", size = 0.5) +
  labs(x = "Individual",
    y = "Score") +
  theme_minimal()
```

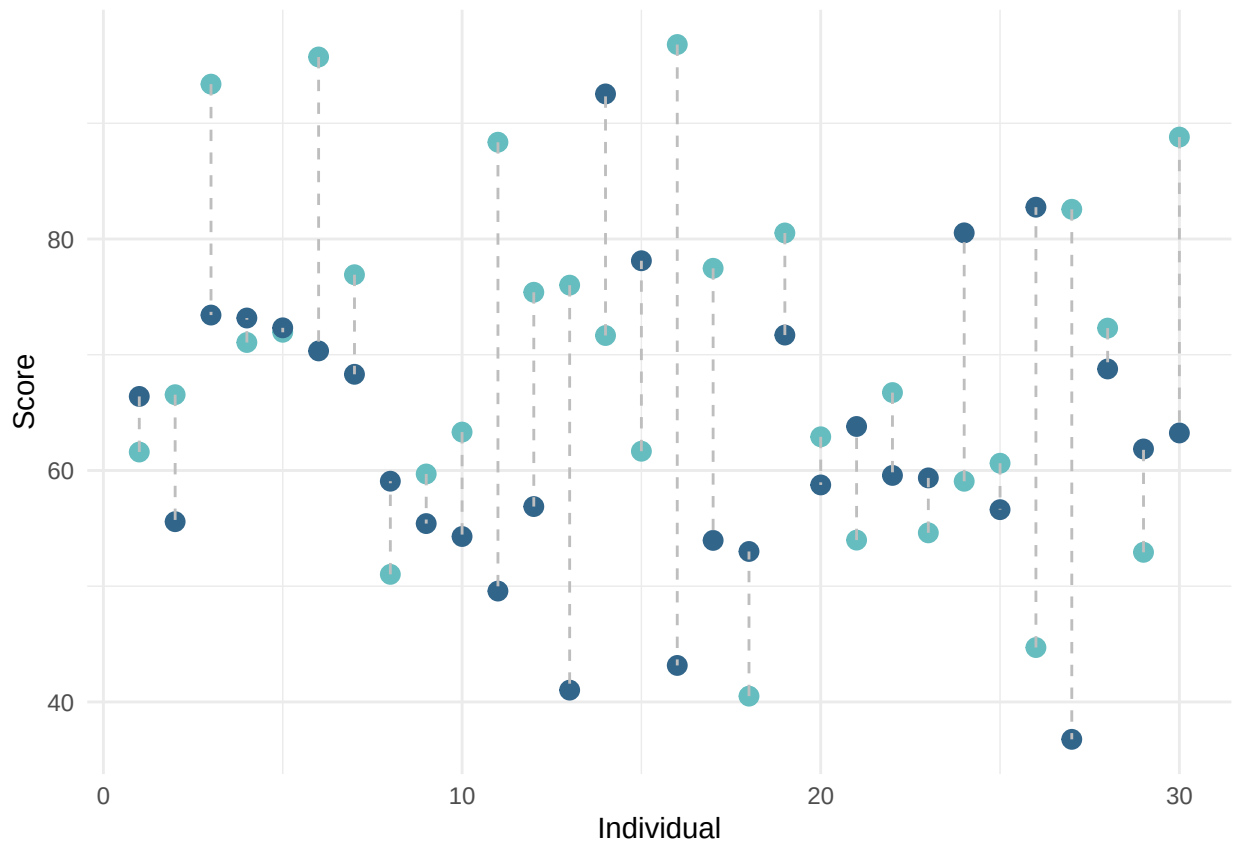


Figure 5: ADHD scores in individuals pre- and post-treatment. Lines connect observations from a given individual.

```
#ggsave("connected_dot_plot.pdf", width = 4, height = 4)
```

Visually, nothing is very obvious, but potentially post-treatment scores are lower.

5.4 Statistical Test

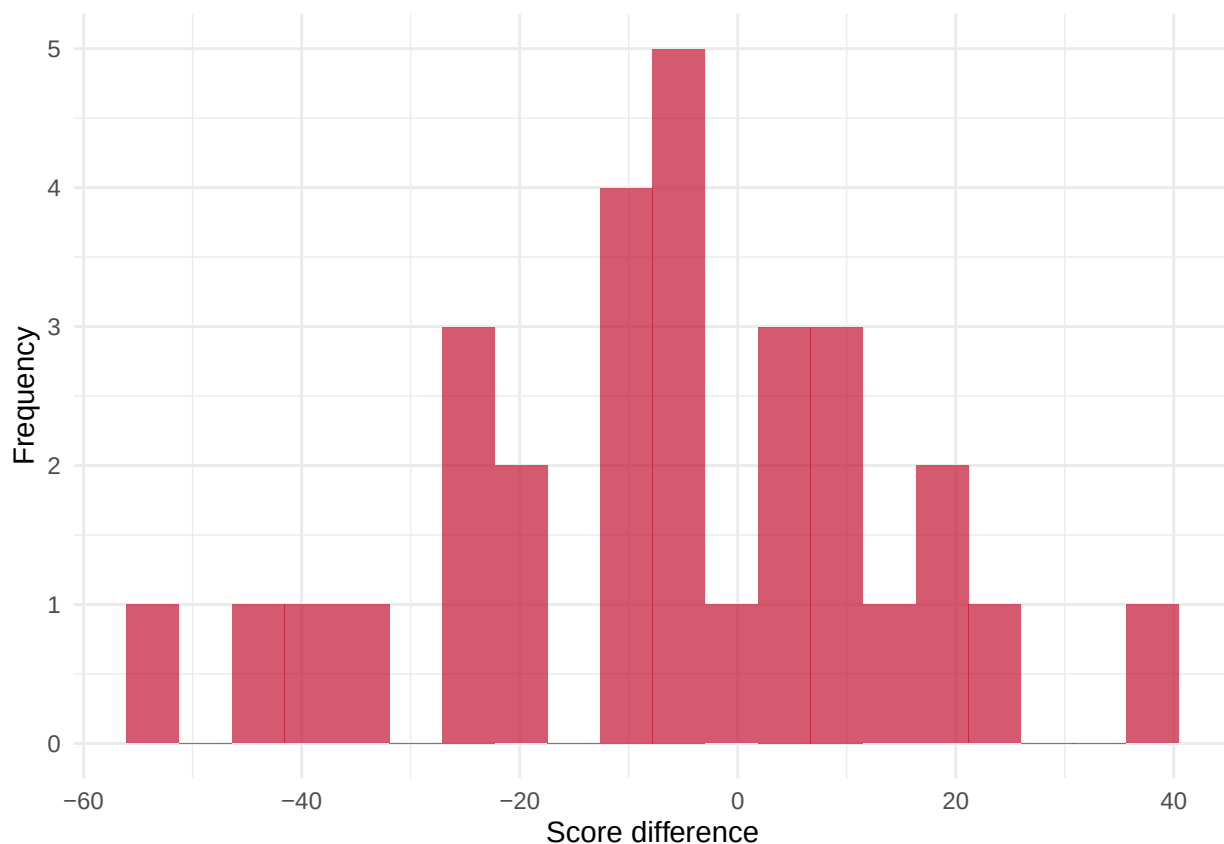
5.4.1 Assumptions

5.4.1.1 Independence *Observations from each population are independent of one another.* This is likely the case (would be determined by the study design).

5.4.1.2 Normality *The within-pair difference is normally distributed.* Here we need to look at the distribution of the **within-pair difference**.

```
# Create new variable
data_adhd$within_pair_difference = data_adhd$Post - data_adhd$Pre

# Create a histogram with different colors for pre- and post-treatment
ggplot(data_adhd, aes(x = within_pair_difference)) +
  geom_histogram(position = "identity", alpha = 0.7, bins = 20, fill = "#C31230") +
  labs(x = "Score difference",
       y = "Frequency") +
  theme_minimal()
```



```
#ggsave("histo_diff.pdf", width = 4, height = 2)
```

Note that now, the difference looks slightly below 0, so maybe there is a difference between the pre- and post-scores within pairs!

It looks normal enough.

5.4.2 Test

```
t.test(x = data_adhd$Pre, y = data_adhd$Post, paired = TRUE)

##
## Paired t-test
##
## data: data_adhd$Pre and data_adhd$Post
## t = 1.7451, df = 29, p-value = 0.09156
## alternative hypothesis: true mean difference is not equal to 0
```

```
## 95 percent confidence interval:  
## -1.138395 14.375133  
## sample estimates:  
## mean difference  
## 6.618369
```

We do not reject the null hypothesis, i.e., we do not detect a difference. Output fits with expectations.

5.4.3 Explanation

Using a paired t-test, we did not detected any significant effect of treatment on ADHD score.